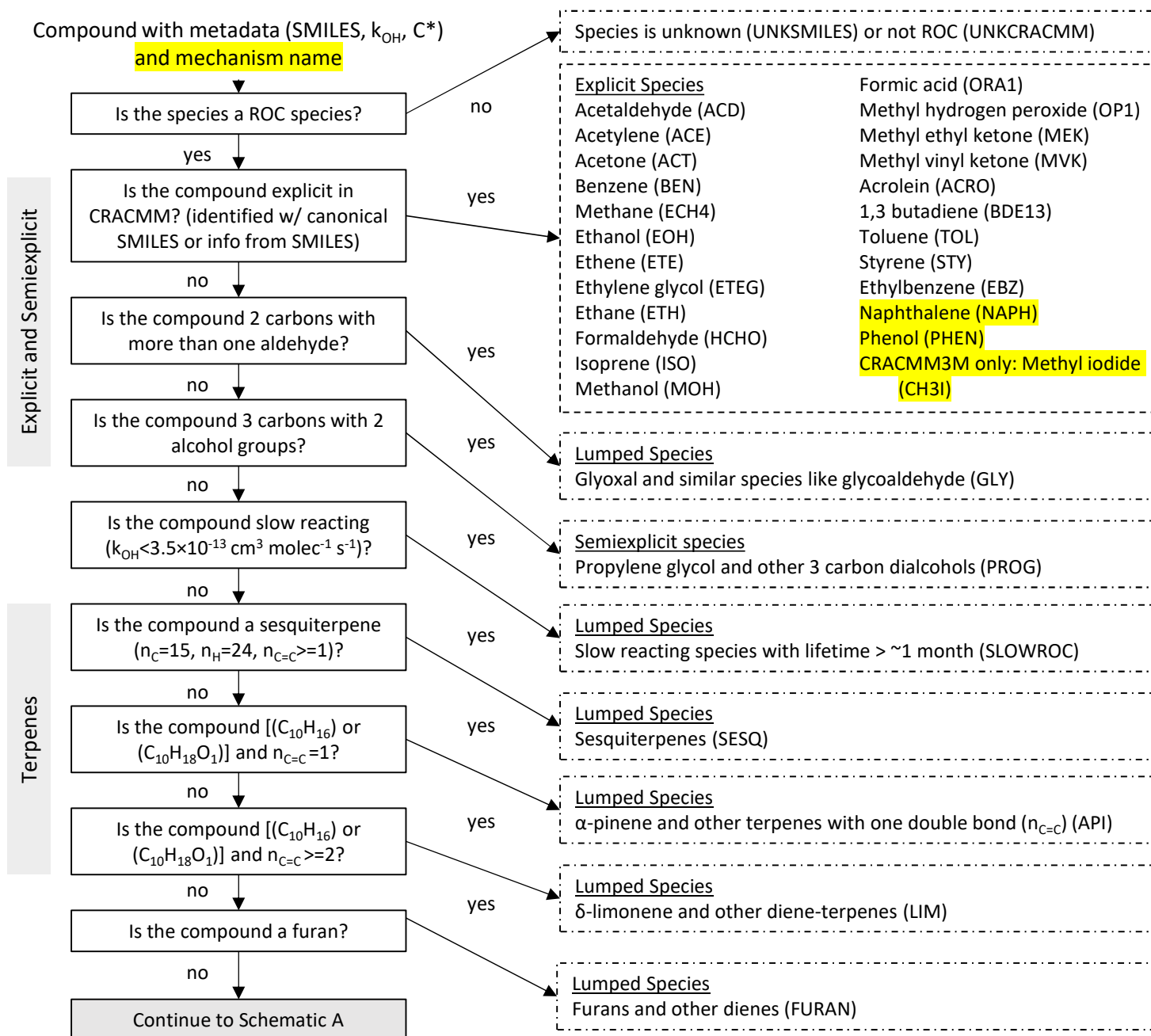


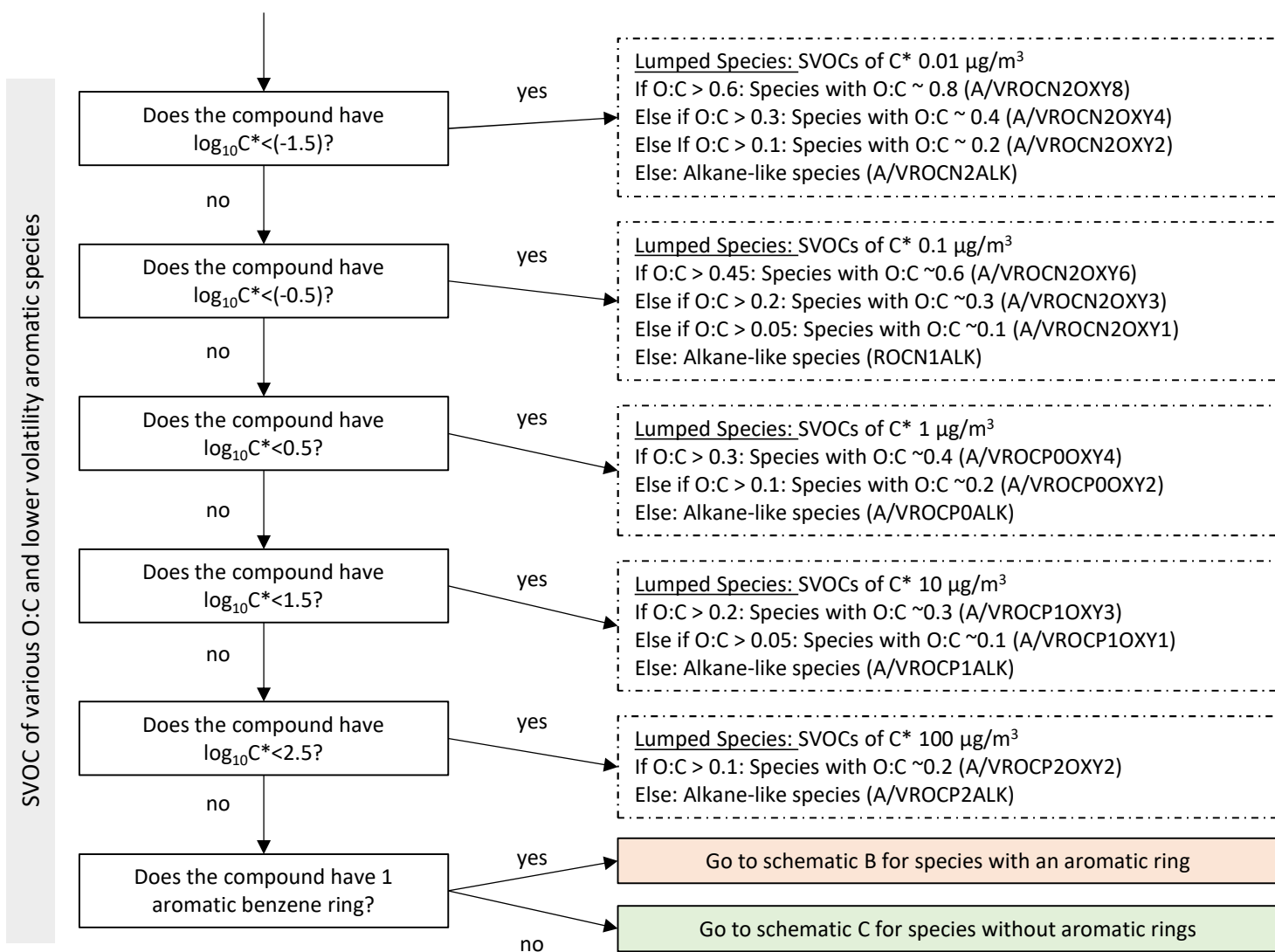
CRACMMv3.0 Emissions Mapping, Updated 6/8/2026



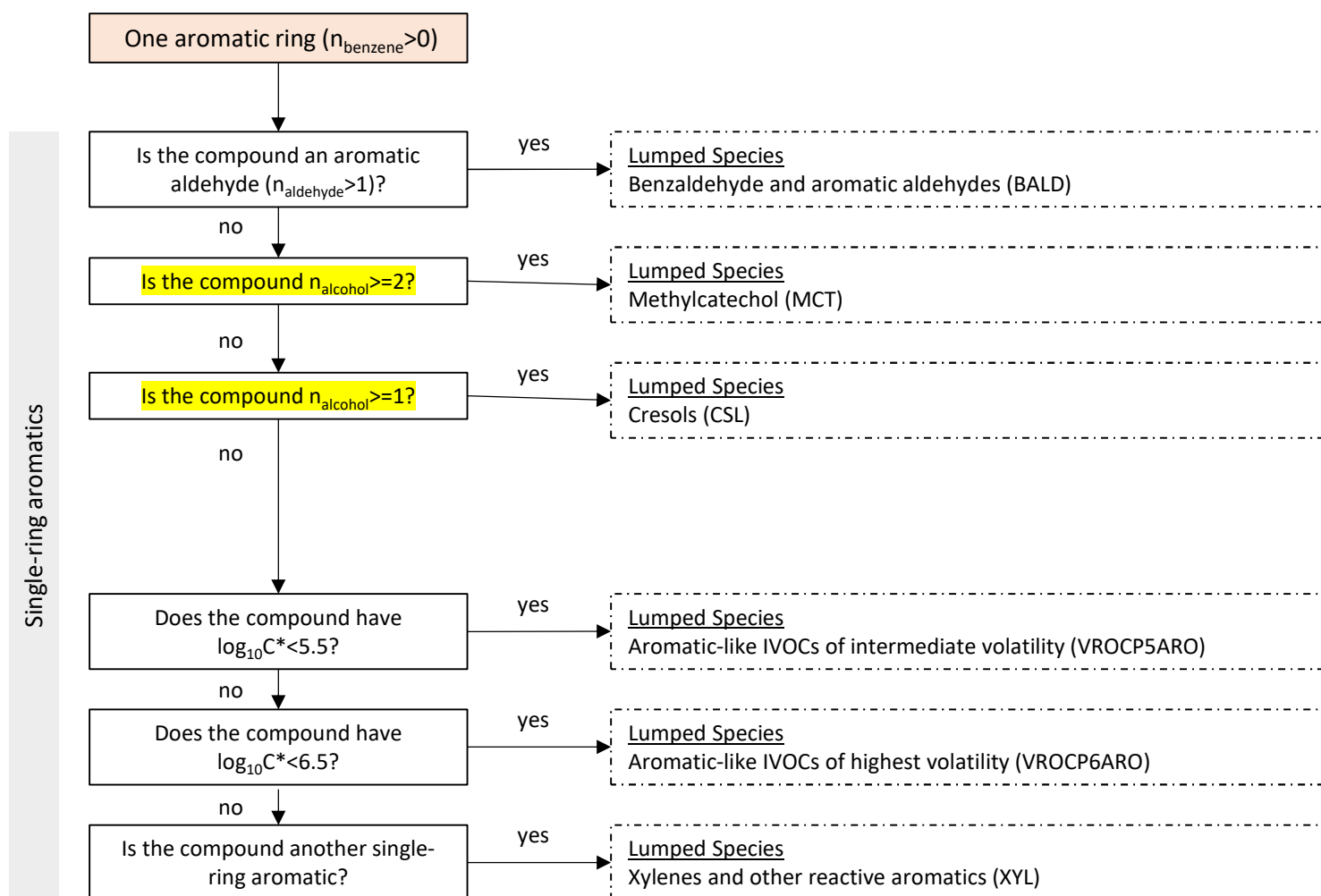
Functional updates to the cracmm mapper for CRACMM3 highlighted in yellow.

The mapper can map to CRACMM3 or CRACMM3M. The only difference in CRACMM3M is explicit CH3I.

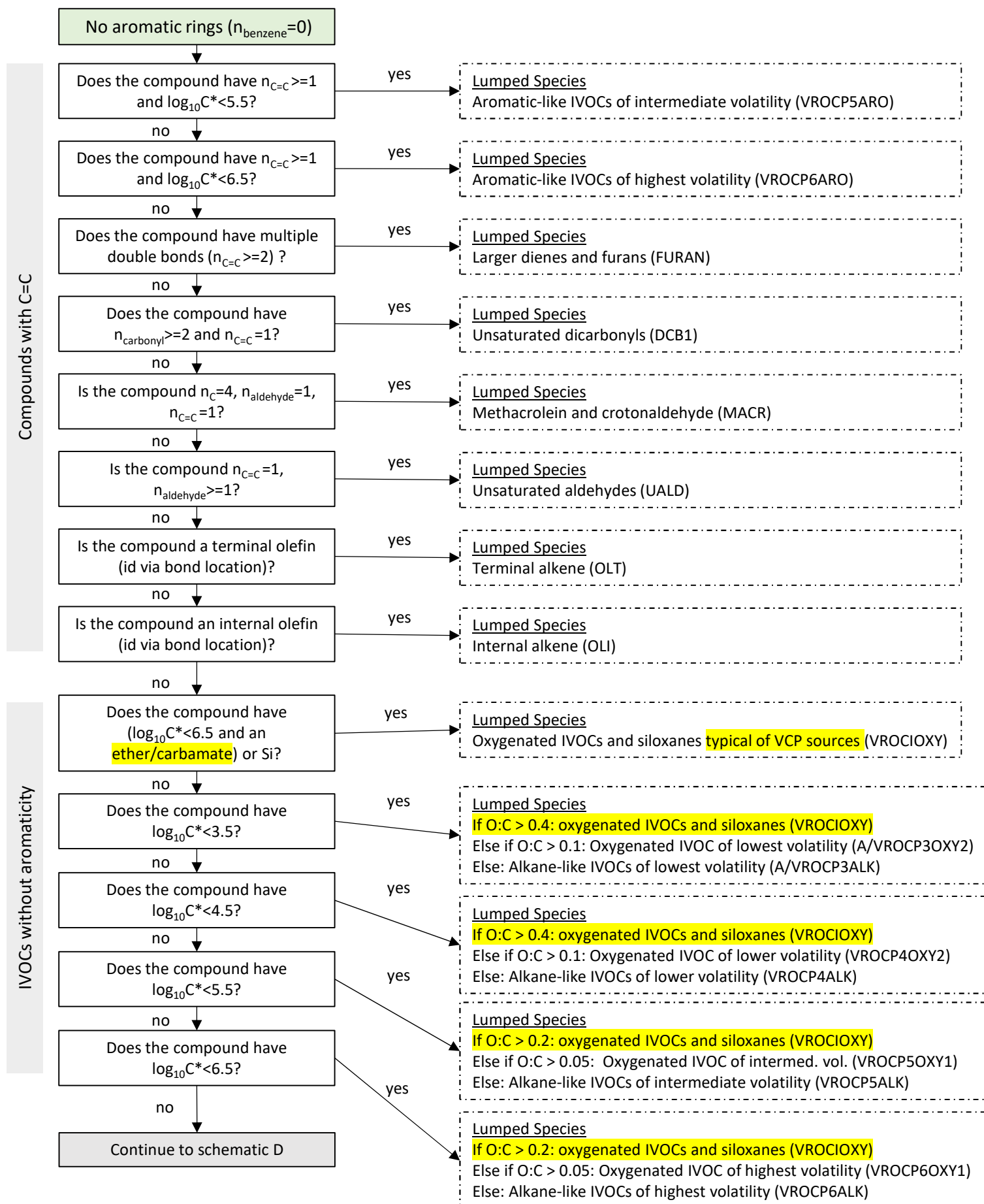
Schematic A



Schematic B

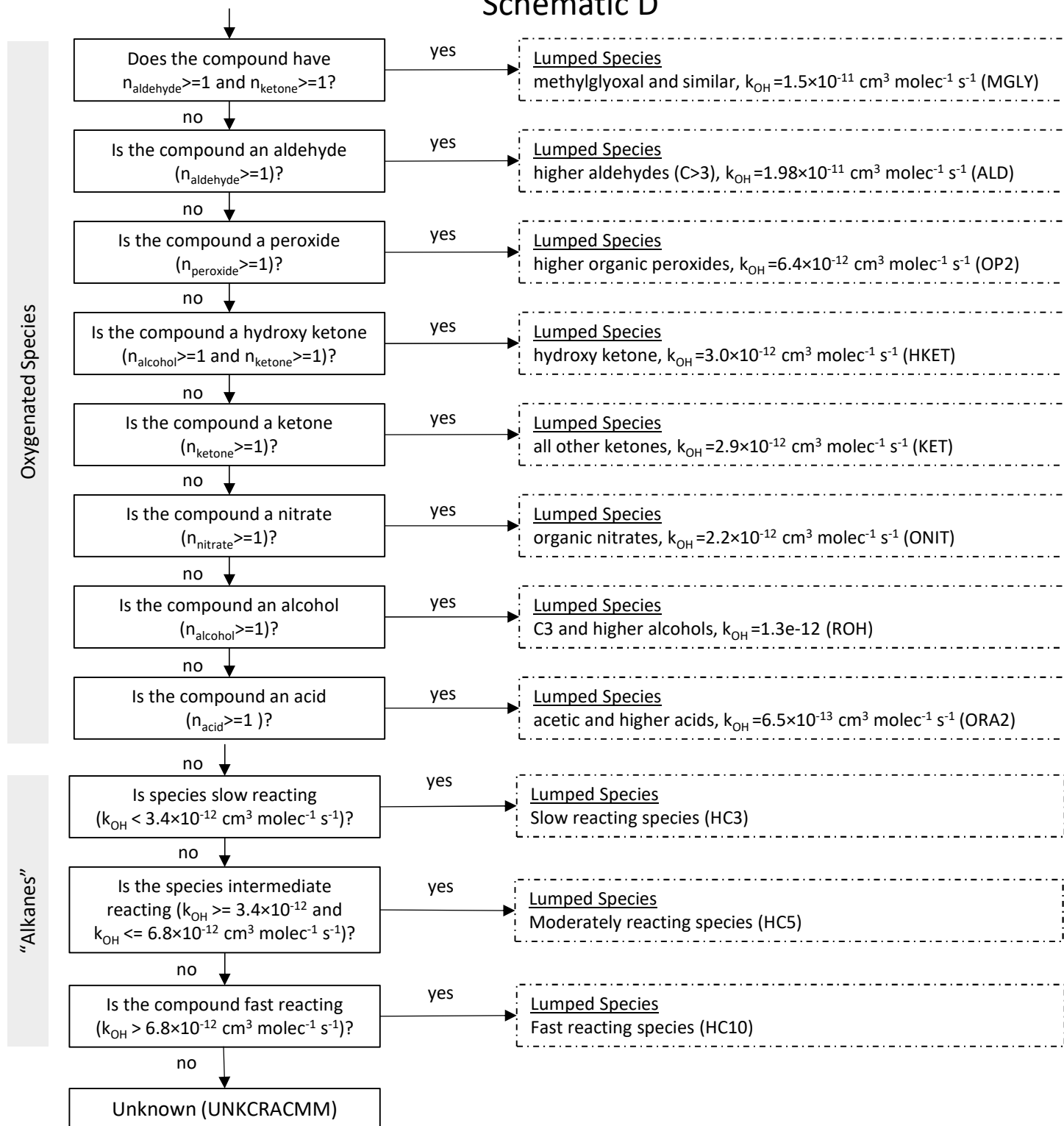


Schematic C



CRACMMv3.0 Emissions Mapping, Updated 6/8/2026

Schematic D



The CRACMM3 emission mapper includes some other updates for robustness, specifically more use of canonical smiles. These updates are not expected to change any previous CRACMM2 mappings. Some species can exist in a gas (V) or aerosol (A) phase. The python mapper includes an optional argument to label a species as gas or particle if two phases are possible. The user must specify the phase as input to the mapper (it is not diagnosed). The phase can be calculated based on C* and organic aerosol concentration outside the mapper.

C* are always in $\mu\text{g m}^{-3}$ in this diagram.

Unknowns may be of 3 types:

- UNKKOH: unknown k_{OH} (correct by specifying better surrogate).
- UNKSMILES: unknown SMILES (correct by specifying better surrogate).
- UNKCRACMM: unknown in mapping. All ROC species eventually get classified by k_{OH} , but species that do not have any carbon atoms or are elemental carbon do not get mapped.

CRACMM2 documentation: Skipper et al. (2024) <https://doi.org/10.5194/acp-24-12903-2024>;

CRACMM1 documentation: Pye et al. (2023) <https://doi.org/10.5194/acp-23-5043-2023> and Place et al. (2023) <https://doi.org/10.5194/acp-23-9173-2023>

RAM2 SI: Goliff et al. (2013) <https://ars.els-cdn.com/content/image/1-s2.0-S1352231012011065-mm1.pdf>